INTERNATIONAL MONETARY FUND AND
INTERNATIONAL DEVELOPMENT ASSOCIATION

THE FEDERAL DEMOCRATIC REPUBLIC OF ETHIOPIA

Joint IMF/World Bank Debt Sustainability Analysis 2007

Prepared by the staffs of the International Monetary Fund and
the International Development Association

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May 22, 2007

I. External Debt Sustainability Analysis

In recent years Ethiopia has benefited from significant debt relief, which has helped to ensure that its debt is sustainable. HIPC debt relief in 2004 and MDRI relief in 2006 lowered Ethiopia's debt stock by about US$6 billion (about 40 percent of GDP), improving its debt sustainability outlook substantially. Indeed, as of the beginning of 2006/07, the net present value (NPV) of debt in relation to output stood at 6 percent and to exports at 36 percent, far below the thresholds suggested by the debt sustainability framework (DSF) developed by the World Bank and the Fund.

The DSA shows Ethiopia at moderate risk of debt distress. While the debt burden indicators are below the policy dependent-thresholds for the whole projection period in the baseline, one stress test reaches the threshold in terms of debt to exports (Figure 1). In the baseline scenario, the NPV of debt to exports is projected to reach about 60 percent in five years and then hold there over the long run; this is well below the prudent threshold (150 percent of exports) set in connection with the DSF for a country classified as a "medium performer" based on the World Bank's Country Policy and Institutional Assessment rating (2003-2005). The NPV of debt to GDP is projected to level out at about 15 percent by 2025/26, also well below the prudent threshold of 40 percent of GDP. Debt service is manageable throughout the period, rising only to 3 percent of exports in 2025/26.

Over the medium term the debt profile based on historical parameters is less favorable in terms of output but comparable to the baseline scenario in terms of exports. The ratio of debt to GDP based on historical patterns is considerably less favorable than the baseline forecast. This is because the average annual growth rate in U.S. dollars for 1995–2005 period was only 5 percent but it is projected to average almost 8 percent over the next two decades. Debt stabilizes at about 70 percent of exports in the historical scenario, slightly higher than in the baseline scenario. This is attributable to the fact that higher debt financing is needed to cover the average current account deficit for the 1995–2005 period of 3.3 percent of GDP (used as the basis for the historical scenario) than for the projected current account deficit of 2 percent for the 2009–25 period.

Box 1. Macroeconomic Assumptions for the Baseline Scenario

Real GDP grows annually at about 8 percent until 2010/11, falling to 7 percent thereafter.

Exports of goods and services are projected to grow in volume at an annual rate of about 9 ½ percent between 2007/08 and 2010/11 but export prices are projected to remain flat. Thereafter, exports grow at 11 ½ percent because prices are assumed to improve.

The current account deficit (before official transfers) is projected to decline from 17 percent of GDP to 12 ½ percent through 2010/11 and to improve further to average out at 7 percent of GDP over the long term. However, official international **reserves coverage** is projected to be at the lower end of the target of 10–12 weeks of imports over the next five years because of high import growth but then rise to 5 months of imports by 2025/26.

Foreign direct investment is envisaged to average about 2 ½ percent of GDP through 2010/11 and rise to 2 ¾ percent of GDP thereafter.

Loan financing on concessional terms is projected to average about 4 percent of GDP annually through 2010/11 but then decline to below 1 percent of GDP by 2025/26.

Sensitivity analysis

For Ethiopia, the two most extreme stress tests involve (i) applying one half standard deviation shock to growth, exports and grants and (ii) new borrowing at less favorable rates. Since Ethiopia has been accustomed to severe droughts over the past two decades, the standard deviation of output and export growth is high. Applying one half standard deviation shocks to growth, exports, and grants in 2006/07 and 2007/08 has a very important bearing on the debt profile since the ratio of debt to exports peaks at about 155 percent in 2010/11 and eventually levels out at 95 percent in 2025/26.

While borrowing at less favorable rates raises the level of debt in relation to exports by about 50 percent, it does not lead to a breach in the debt threshold. Over half of the new lending over the medium term is disbursed by the World Bank at an interest rate of 0.75 percent with a 10-year grace period and a 50-year repayment period. The stress test involving more expensive financing assumes that the average interest rate is higher by 2 percentage points for the whole forecast period. Since Ethiopia is expected to continue to borrow large amounts over the medium-term, this interest rate hike has significant adverse effects on the debt profile. However, the effects are much more gradual than for the scenario with shocks with the debt to export ratio peaking at 107 percent in 2018/19. Higher financing costs raise the debt-to-GDP ratio to peak at about 25 percent, and the debt service-to-export ratio levels out at 5 ½ percent. All these estimates are below the prudent thresholds.

The debt profile is sensitive to assumptions made about export growth in the long run. The baseline long-run forecast assumes that export growth levels out at about 11 ½ percent a

year. This estimate is comparable to the historical average for 1995–2005. However, it also implies a rising export share of output over time, since output in U.S. dollars is projected to grow at 8 percent a year. If the export ratio is projected to stabilize over the forecast period, this would have adverse effects on the debt to export ratio. Figure 2 presents the baseline projection and the profile of debt and debt service in relation to exports, assuming that exports in US dollars grow at the rate of output. In this case debt rises to about 100 percent of exports by 2025/26 in the sensitivity scenario, while debt service in relation to exports stabilizes at about 5 ½ percent a year. Though these figures are considerably higher than in the baseline scenario, they are still below the prudent thresholds.

The inclusion of commercial loans to public enterprises has a sizeable effect on the DSA over the medium-term and is another justification for categorizing the debt risk as moderate. The baseline scenario excludes commercial loans to public enterprises because they are not publicly guaranteed. This is one reason why Ethiopian Airline debt was not included in the ceiling of noncommercial debt in recent Fund-supported programs with Ethiopia. Ethiopian Airlines is currently financing 10 new Boeing 787 airplanes through 2012 at a cost of US$1.2 billion and the Ethiopian electricity company also has ambitious investment plans. The Ethiopian telecom company is also investing up to US$1.5 billion over the next 10 years. Incorporating these loans into the baseline (Figure 2), the NPV of debt to exports peaks at 140 percent in 2011/12 but then settles down to 60 percent over the long term. The debt service to exports ratio rises rapidly to peak at 14 percent within five years but then gradually falls toward the baseline estimate as the loans of the public enterprises are paid off. The stress tests also show considerably higher debt burden ratios when public enterprise debt is included, with breaches of all three indicative thresholds.

I. Fiscal Sustainability Analysis

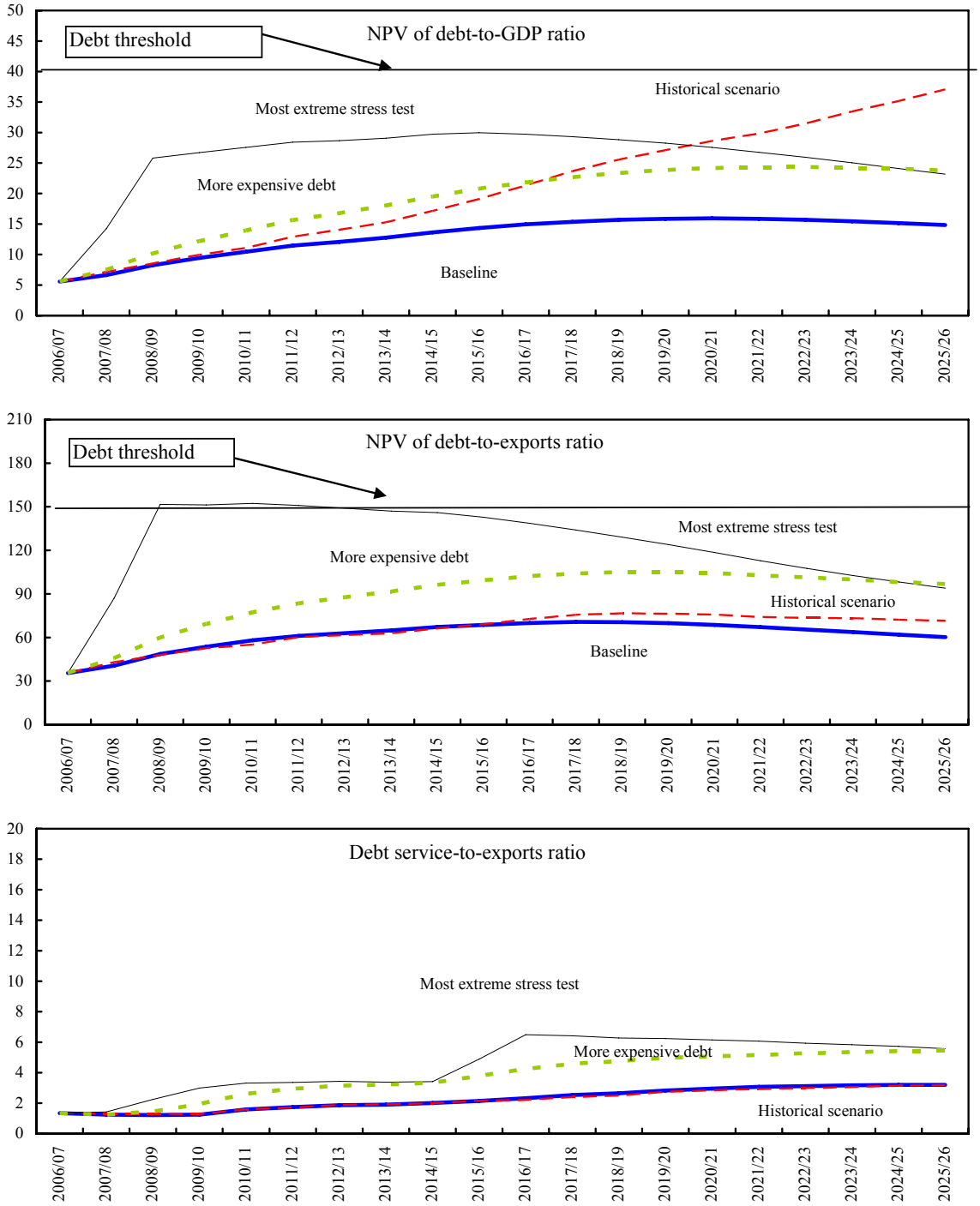
The fiscal DSA is based on the assumption that annual domestic financing over the medium term will be around 2–2 ½ percent of GDP. As a result, the ratio of total debt to GDP is projected to decline gradually from about 40 percent of GDP in 2005/06 to below 30 percent in the long run (Figure 3). This profile is slightly stronger than the corresponding scenario based on historical averages for the main macroeconomic determinants of public debt dynamics. One reason for the rapid decline in total debt over the medium term is that domestic interest rates rise only gradually to level out at 7 percent. If we assume a 3 percentage point increase in interest rates over the complete projection period, the ratio of total debt to GDP falls only modestly and stays at around 35 percent in the long run.

II. Conclusion

While external debt ratios have improved considerably with debt relief, the DSA shows a moderate risk of debt distress based on stress tests. These stress tests capture the vulnerability of the Ethiopian economy to shocks and illustrate the importance of securing grant and concessional financing and containing domestic borrowing if debt is to stay

sustainable over the medium to long term. Also important are strengthened debt management capacity and a comprehensive public debt strategy that specifically includes public enterprise debt and contingent liabilities. Finally, the impact on debt ratios of new borrowing in infrastructure sectors underlines the need to ensure that these investments lead to sufficiently high returns.

Figure 1. Ethiopia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2006-2025,
(In percent)



Source: Annex II, Tables 1 and 2.

Notes: The most extreme stress test involves one half standard deviation shocks to output and export growth and grants. In the debt service-to-exports panel, the debt service threshold is 20 percent

Figure 2. Ethiopia: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2006-2025,
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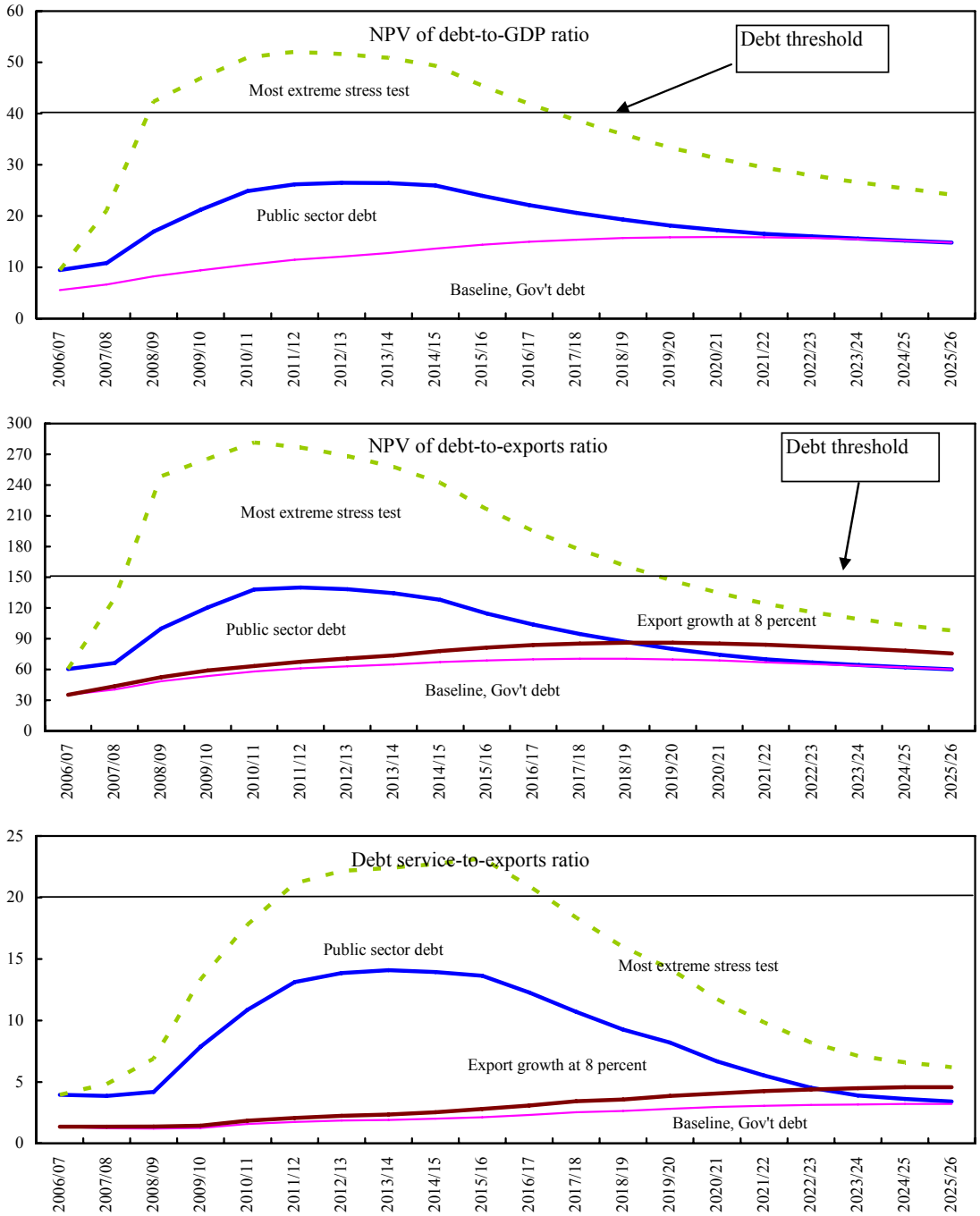
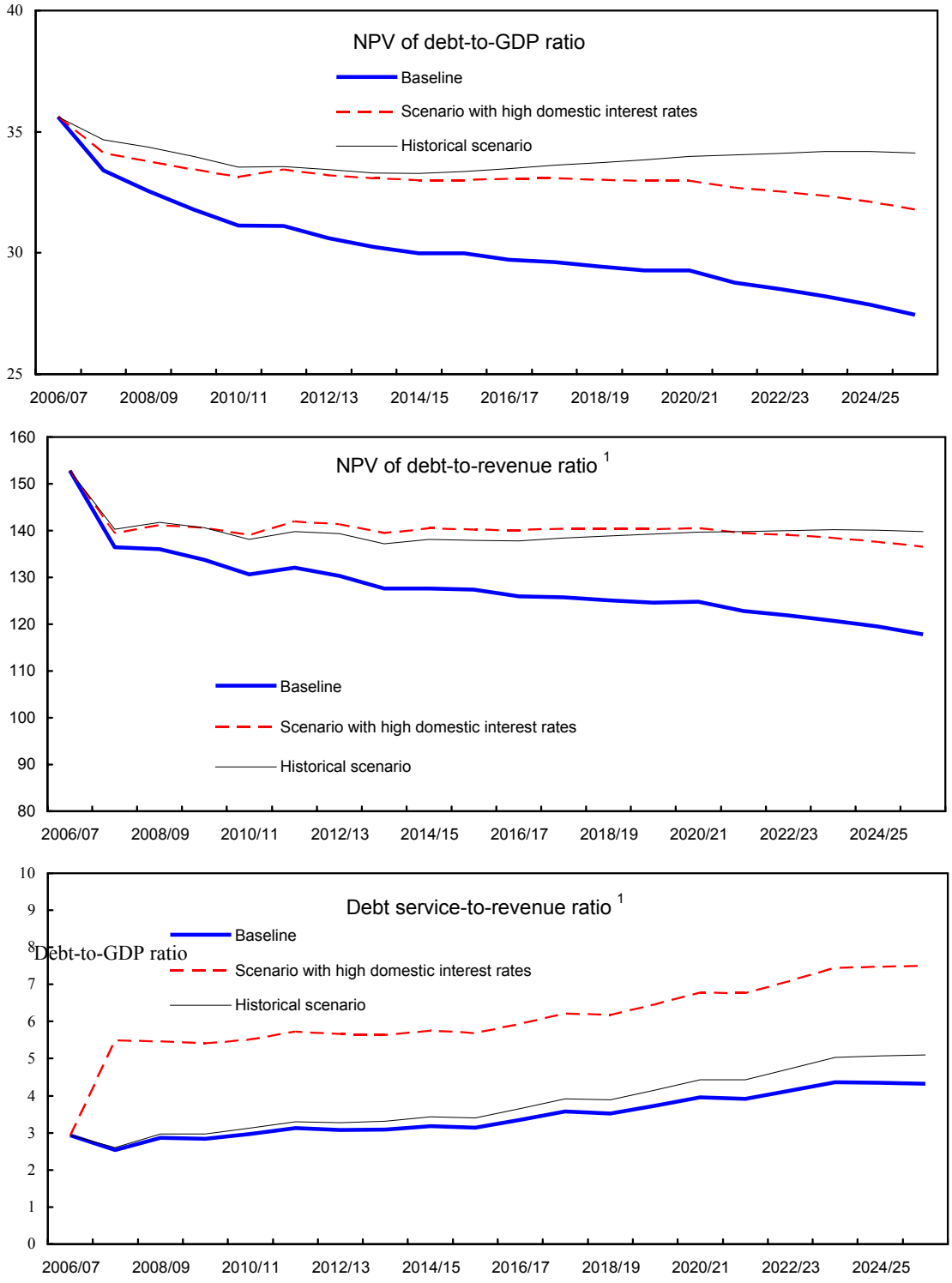


Source: Annex II, Tables 1 and 2.

Notes: In the debt service-to-exports panel, the debt service threshold is 20 percent

Figure 3. Indicators of Public Debt, 2005/06-2025/26



Source: IMF staff projections and simulations.
[1] Revenue including grants.

Table 1. Ethiopia: External Debt Sustainability Framework, Baseline Scenario, 2006-2026 1/

(In percent of GDP, unless otherwise indicated)

	Projections							
	2006/07	2007/08	2008/09	2009/10	2010/11	2015/16	2020/21	2025/26
External debt (nominal) 1/	**13.1**	**14.9**	**17.9**	**20.0**	**21.7**	**27.4**	**28.6**	**25.9**
o/w public and publicly guaranteed (PPG)	13.1	14.9	17.9	20.0	21.7	27.4	28.6	25.9
Change in external debt	-31.7	1.8	3.0	2.1	1.7	1.0	-0.1	-0.7
Identified net debt-creating flows	-0.9	0.1	1.7	0.5	0.5	-1.8	-2.6	-4.6
Non-interest current account deficit	**5.5**	**3.7**	**5.1**	**4.1**	**4.2**	**2.4**	**1.9**	**-0.4**
Deficit in balance of goods and services	22.6	21.7	22.3	21.7	22.1	19.2	19.1	17.1
Exports	15.7	16.4	17.0	17.6	18.0	20.9	23.2	24.6
Imports	38.3	38.1	39.3	39.3	40.2	40.1	42.3	41.7
Net current transfers (negative = inflow)	-16.9	-17.9	-17.0	-17.6	-18.0	-16.8	-16.5	-16.5
Other current account flows (negative = net inflow)	-0.2	-0.1	-0.1	0.0	0.1	0.1	-0.7	-1.0
Net FDI (negative = inflow)	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-2.5**	**-2.8**	**-2.8**	**-2.8**
Endogenous debt dynamics 2/	**-3.9**	**-1.0**	**-0.9**	**-1.1**	**-1.1**	**-1.4**	**-1.6**	**-1.5**
Contribution from nominal interest rate	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2
Contribution from real GDP growth	-4.0	-1.1	-1.0	-1.2	-1.3	-1.6	-1.8	-1.7
Contribution from price and exchange rate changes	…	…	…	…	…	…	…	…
Residual (3-4) 3/	**-30.8**	**1.7**	**1.3**	**1.6**	**1.2**	**2.8**	**2.4**	**3.9**
o/w exceptional financing	-0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0
NPV of external debt 4/	9.5	10.8	17.0	21.2	24.9	24.0	17.3	14.8
In percent of exports	60.6	66.3	100.0	120.6	138.1	114.5	74.5	60.3
NPV of PPG external debt	**5.6**	**6.6**	**8.3**	**9.4**	**10.5**	**14.4**	**15.9**	**14.8**
In percent of exports	35.5	40.6	48.7	53.7	58.1	68.7	68.7	60.3
Debt service-to-exports ratio (in percent)	1.4	1.2	1.2	1.3	1.6	2.1	3.0	3.2
PPG debt service-to-exports ratio (in percent)	1.4	1.2	1.2	1.3	1.6	2.1	3.0	3.2
Total gross financing need (billions of U.S. dollars)	0.5	0.2	0.5	0.4	0.4	0.0	-0.1	-1.7
Non-interest current account deficit that stabilizes debt ratio	37.3	1.9	2.1	2.0	2.5	1.5	2.0	0.3
Key macroeconomic assumptions								
Real GDP growth (in percent)	10.7	9.6	7.7	7.3	6.8	6.7	6.9	6.9
GDP deflator in US dollar terms (change in percent)	7.9	2.3	1.9	1.1	0.5	1.1	1.1	1.5
Effective interest rate (percent) 5/	0.3	0.9	0.9	0.8	0.9	0.8	0.8	0.8
Growth of exports of G&S (US dollar terms, in percent)	18.3	17.1	13.8	12.4	10.0	11.2	10.4	9.0
Growth of imports of G&S (US dollar terms, in percent)	9.7	11.6	13.0	8.6	9.7	8.1	9.0	7.0
Grant element of new public sector borrowing (in percent)	53.4	50.8	52.3	53.4	52.5	50.7	51.6	52.4
Memorandym item:								
Nominal GDP (billions of US dollars)	15.9	17.8	19.6	21.2	22.8	33.2	48.8	73.1

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Ethiopia: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2006-26
(percent of GDP unless otherwise stated)

	Projections							
	2006/07	2007/08	2008/09	2009/10	2010/11	2015/16	2020/21	2025/26
NPV of debt-to-GDP ratio								
Baseline	5.6	6.6	8.3	9.4	10.5	14.4	15.9	14.8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	5.6	7.1	8.5	9.9	11.0	19.0	28.6	37.2
A2. New public sector loans on less favorable terms in 2007-26 2/	5.6	7.6	10.3	12.4	14.1	21.1	24.7	24.3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	5.6	7.3	9.9	11.3	12.6	17.4	19.3	18.0
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	5.6	8.2	12.5	13.5	14.4	17.6	18.0	16.1
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	5.6	7.4	10.1	11.5	12.8	17.6	19.6	18.2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	5.6	13.0	20.3	21.0	21.8	23.8	22.0	18.5
B5. Combination of B1-B4 using one-half standard deviation shocks	5.6	14.4	26.3	27.2	28.1	30.5	28.0	23.5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	5.6	9.5	11.8	13.5	15.0	20.7	23.0	21.4
NPV of debt-to-exports ratio								
Baseline	35.5	40.6	48.7	53.7	58.1	68.7	68.7	60.3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	35.1	42.2	47.2	51.7	54.2	68.2	74.8	70.9
A2. New public sector loans on less favorable terms in 2007-26 2/	35.9	46.6	60.9	70.4	78.5	101.1	106.6	98.8
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	35.5	40.0	48.1	53.1	57.5	68.3	68.5	60.5
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	35.5	62.0	110.7	115.5	120.2	126.7	116.7	98.8
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	35.5	40.0	48.1	53.1	57.5	68.3	68.5	60.5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	35.5	78.0	117.1	117.0	118.0	111.1	92.7	74.0
B5. Combination of B1-B4 using one-half standard deviation shocks	35.5	88.3	154.5	154.1	155.1	145.1	120.3	95.6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	35.5	40.0	48.1	53.1	57.5	68.3	68.5	60.5
Debt service ratio								
Baseline	1.4	1.2	1.2	1.3	1.6	2.1	3.0	3.2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	1.4	1.3	1.3	1.3	1.6	2.1	2.8	3.2
A2. New public sector loans on less favorable terms in 2007-26 2/	1.4	1.3	1.5	2.0	2.6	3.8	5.1	5.5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	1.4	1.2	1.2	1.2	1.6	2.1	2.9	3.2
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	1.4	1.6	2.1	2.5	3.0	4.0	5.5	5.5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	1.4	1.2	1.2	1.2	1.6	2.1	2.9	3.2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	1.4	1.2	1.8	2.3	2.6	3.9	4.8	4.4
B5. Combination of B1-B4 using one-half standard deviation shocks	1.4	1.4	2.3	3.1	3.4	5.0	6.3	5.7
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	1.4	1.2	1.2	1.2	1.6	2.1	2.9	3.2
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	51	51.3	51.3	51.3	51.3	51.3	51.3	51.3

Source: IMF staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.

2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.

3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3.Country: Public Sector Debt Sustainability Framework, 2005/06-2025/26 (Baseline Scenario)
(Percent of GDP, unless otherwise indicated)

	Projections								
	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2015/16	2020/21	2025/26
Public sector debt [1]	43.2	41.7	42.3	42.4	42.4	43.0	42.8	42.2	38.6
Of which : foreign-currency denominated	13.1	15.0	18.0	20.1	21.8	23.4	26.5	28.8	25.9
Change in public sector debt	-36.7	-1.5	0.6	0.2	0.0	0.6	0.6	-0.3	-0.8
Identified debt-creating flows	-34.9	-3.0	-2.0	-1.5	-0.9	-0.4	-0.7	-0.6	-0.4
Primary deficit	3.6	3.2	3.3	3.2	3.1	3.3	2.9	2.7	2.5
Revenue and grants	23.3	24.5	23.9	23.8	23.8	23.6	23.5	23.5	23.3
Of which : grants	6.0	6.2	5.7	5.7	5.7	5.7	5.5	5.2	4.6
Primary (noninterest) expenditure	26.9	27.6	27.2	27.0	27.0	26.8	26.4	26.2	25.8
Automatic debt dynamics	-13.9	-6.1	-5.3	-4.7	-4.0	-3.7	-3.5	-3.3	-2.9
Contribution from interest rate/growth differential	-14.4	-7.0	-6.7	-6.4	-5.8	-5.5	-5.5	-5.4	-4.9
Of which : contribution from average real interest rate	-7.5	-3.6	-3.7	-3.5	-3.1	-3.0	-2.8	-2.7	-2.3
Of which : contribution from real GDP growth	-6.8	-3.4	-3.0	-2.9	-2.7	-2.5	-2.6	-2.7	-2.6
Contribution from real exchange rate depreciation	0.5	0.9	1.4	1.6	1.8	1.8	1.9	2.2	...
Other identified debt-creating flows	-24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	-24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	-1.8	1.5	2.5	1.6	0.9	1.0	1.3	0.3	-0.4
NPV of public sector debt	35.6	33.4	32.6	31.8	31.1	31.1	30.0	29.3	27.5
Of which : external	5.6	6.7	8.3	9.5	10.5	11.5	13.7	15.9	14.8
Gross financing need [2]	4.4	3.9	4.1	4.0	3.9	4.2	3.8	4.0	4.1
NPV of public sector debt-to-revenue ratio (percent) [3]	152.8	136.5	136.0	133.7	130.6	132.1	127.7	124.6	117.9
o/w external	23.9	27.2	34.5	39.8	44.0	48.7	58.1	67.6	63.7
Debt service-to-revenue ratio (percent) [3 4]	3.6	3.0	3.2	3.2	3.5	3.8	4.1	4.7	6.8
Primary deficit that stabilizes the debt-to-GDP ratio	40.3	4.7	2.7	3.1	3.1	2.7	2.3	3.0	3.3
Key macroeconomic and fiscal assumptions									
Real GDP growth (percent)	9.4	8.5	7.7	7.3	6.8	6.3	6.6	6.9	7.2
Average nominal interest rate on forex debt (percent)	1.0	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Average real interest rate on domestic currency debt (percent)	-11.9	-8.9	-8.6	-7.7	-6.4	-6.0	-7.7	-5.4	-3.3
Inflation rate (GDP deflator, percent)	16.0	12.6	12.3	11.4	10.2	9.8	9.8	9.8	9.8
Growth of real primary spending (deflated by GDP deflator, percent)	20.5	12.9	6.1	6.3	6.4	5.6	5.4	6.6	7.0

Sources: Ethiopian authorities; and IMF staff estimates and projections.

[1] Covering public and publicly guaranteed debt on gross basis.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues including grants.

[4] Debt service is defined as the sum of interest and amortization of medium- and long-term debt.

Table 4. Country: Public Sector Debt Sustainability Framework, 2005/06-2025/26 (Alternative Scenario with High Interest Rates)
(In percent of GDP, unless otherwise indicated)

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	Projections								
	2006/07	2007/08	2008/09	2009/10	2010/11	2011/12	2015/16	2020/21	2025/26
Public sector debt [1]	43.2	42.4	43.5	44.1	44.4	45.4	45.8	45.9	42.9
Of which : foreign-currency denominated	13.1	15.0	18.0	20.1	21.8	23.4	26.5	28.8	25.9
Change in public sector debt	-36.7	-0.8	1.1	0.6	0.4	1.0	0.8	-0.2	-0.7
Identified debt-creating flows	-34.9	-2.2	-1.5	-1.1	-0.5	-0.1	-0.5	-0.5	-0.3
Primary deficit	3.6	3.2	3.3	3.2	3.1	3.3	2.9	2.7	2.5
Revenue and grants	23.3	24.5	23.9	23.8	23.8	23.6	23.5	23.5	23.3
Of which : grants	6.0	6.2	5.7	5.7	5.7	5.7	5.5	5.2	4.6
Primary (noninterest) expenditure	26.9	27.6	27.2	27.0	27.0	26.8	26.4	26.2	25.8
Automatic debt dynamics	-13.9	-5.4	-4.8	-4.3	-3.6	-3.4	-3.3	-3.2	-2.8
Contribution from interest rate/growth differential	-14.4	-6.3	-6.2	-5.9	-5.4	-5.2	-5.3	-5.3	-4.8
Of which : contribution from average real interest rate	-7.5	-2.9	-3.1	-3.0	-2.6	-2.6	-2.5	-2.4	-1.9
Of which : contribution from real GDP growth	-6.8	-3.4	-3.0	-3.0	-2.8	-2.6	-2.8	-3.0	-2.9
Contribution from real exchange rate depreciation	0.5	0.9	1.4	1.6	1.8	1.8	1.9	2.2	...
Other identified debt-creating flows	-24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Debt relief (HIPC and other)	-24.6	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Residual, including asset changes	-1.8	1.4	2.5	1.6	0.9	1.0	1.3	0.3	-0.4
NPV of public sector debt	35.6	34.1	33.8	33.4	33.1	33.5	33.0	33.0	31.8
Of which : external	5.6	6.7	8.3	9.5	10.5	11.5	13.7	15.9	14.8
Gross financing need [2]	4.4	4.6	4.7	4.6	4.6	4.8	4.4	4.6	4.9
NPV of public sector debt-to-revenue ratio (percent) [3]	152.8	139.4	141.1	140.6	139.0	142.0	140.5	140.3	136.5
o/w external	23.9	27.2	34.5	39.8	44.0	48.7	58.1	67.6	63.7
Debt service-to-revenue ratio (percent) [3 4]	3.6	5.9	5.8	5.7	6.1	6.4	6.7	7.0	10.0
Primary deficit that stabilizes the debt-to-GDP ratio	40.3	3.9	2.2	2.7	2.7	2.3	2.1	2.9	3.2
Key macroeconomic and fiscal assumptions									
Real GDP growth (percent)	9.4	8.5	7.7	7.3	6.8	6.3	6.6	6.9	7.2
Average nominal interest rate on forex debt (percent)	1.0	0.9	0.9	0.9	0.9	0.9	0.9	0.9	0.9
Average real interest rate on domestic currency debt (percent)	-9.3	-6.6	-6.2	-5.2	-3.8	-3.3	-5.2	-2.6	-0.4
Inflation rate (GDP deflator, percent)	16.0	12.6	12.3	11.4	10.2	9.8	9.8	9.8	9.8
Growth of real primary spending (deflated by GDP deflator, percent)	20.5	12.9	6.1	6.3	6.4	5.6	5.4	6.6	7.0

Sources: Ethiopian authorities; and IMF staff estimates and projections.

[1] Covering public and publicly guaranteed debt on gross basis.

[2] Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

[3] Revenues including grants.

[4] Debt service is defined as the sum of interest and amortization of medium- and long-term debt.